



SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL
SCIENCES



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE NAME: STATISTICS WITH R PROGRAMMING

COURSE CODE: ITA0404

COURSE OUTCOME COVERED

CO1 -- ASSESSMENT TOOL 2

CO1: Apply R Programming fundamentals including data structures, vectors, matrices, arrays and class types to perform mathematical operations and manage data effectively. (BL: 3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks: 50

Duration: 1 Hour

Pseudocode Writing Task (Algorithm Design) Level2

Code of Conduct:

I B. Vyshnavi Reddy (192525159) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Reg. No: 192525159

RUBRICS FOR CALCULATION

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Q5 (10)	Total Marks (50)
Conceptual Understanding	30						
Accuracy of Answer	25						
Application of Knowledge	20						

Completeness of Response	15						
Clarity & Presentation	10						
Total per Question	10 Marks	/10	/10	/10	/10	/10	/ 50

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1AT-2

NAME: B.Vyshnavi Reddy

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REG.NO: 192525159

COURSE CODE: ITA0404

COURSE NAME: Statistics with R Programming

1.

Aim: To create an employee data frame, identify employees earning more than ₹50,000, store their details in a list, and display the number of eligible employees.

Pseudo Code:

1. Start.
2. Create vectors for Employee ID, Name, Department, and Salary.
3. Create a data frame using the vectors.
4. Create a logical vector where Salary > 50000.
5. Filter the employee data frame using the logical vector.
6. Store the filtered employee details in a list.
7. Count the number of eligible employees.
8. Display the filtered employee list.
9. Display the number of eligible employees.
10. Stop.

Input:

Employee ID
Employee Name
Department Salary

Output:

Employee details earning more than ₹50,000.
Number of eligible employees.

Result: The employee records with salary above ₹50,000 were successfully identified and displayed.

2.

Aim: To create an inventory data frame, calculate the total value of each item, identify lowstock items, and display them.

Pseudo Code:

1. Start.

2. Create vectors for Item Name, Quantity, and Unit Price.
3. Create a data frame using the vectors.
4. Calculate Total Value = Quantity $\times$ Unit Price.
5. Create a logical vector for Quantity less than the minimum stock level.
6. Filter the low-stock items.
7. Store the low-stock item details in a list.
8. Display the low-stock items.
9. Stop.

Input:

ItemName
Quantity
Unit Price

Output:

Total value of each item.
List of low-stock items.

Result:

The inventory was analyzed successfully, and low-stock items were identified and displayed.

3.

Aim: To process student marks, assign grades, create a data frame, and display the number of students in each grade category.

Pseudo Code:

1. Start.
2. Create vectors for student names and marks.
3. Define a function to assign grades based on marks.
4. Apply the function to all marks.
5. Store the grades as a factor.
6. Create a data frame containing Name, Marks, and Grade.
7. Count the number of students in each grade.
8. Display the data frame.
9. Display the grade counts.
10. Stop.

Input:

Student Name Marks

Output:

Student Name, Marks, and Grade.

Number of students in each grade category.

Result:

The student grades were successfully assigned, categorized, and displayed.

4.

Aim: To analyze daily temperature data, identify hot days above 35°C, and display the results.

Pseudo Code:

1. Start.
2. Create a vector containing daily temperatures.
3. Create a logical vector for temperatures greater than 35°C.
4. Count the number of hot days.
5. Store the temperatures and logical vector in a list.
6. Display the list.
7. Display the number of hot days.
8. Stop.

Input:

Daily temperatures

Output:

Logical vector indicating hot days.
Number of hot days.

Result:

The hot days with temperatures above 35°C were successfully identified and counted.

5.

Aim: To create a library data frame, identify books with fewer than five copies, and display the titles requiring replenishment.

Pseudo Code:

1. Start.
2. Create vectors for Book ID, Title, and Number of Copies.
3. Create a data frame using the vectors.
4. Create a logical vector where Number of Copies < 5.
5. Extract the titles of books requiring replenishment.
6. Store the titles in a character vector.
7. Display the titles.
8. Stop.

Input:

BookID

BookTitle

Number of Copies

Output:

Titles of books with fewer than five copies.

Result:

The books requiring replenishment were successfully identified and displayed.